



MICHAEL ANAGLI
R WITH DR. G

FACTORS INFLUENCING PERCEIVED
TRANSPORT SAFETY: A STATISTICAL
ANALYSIS USING R PROGRAMMING

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ABSTRACT

Road safety is an important aspect of sustainable transportation because it influences people's willingness to use different modes of transport. Understanding the factors that affect perceived safety can help urban planners and policymakers design transport systems that encourage active and sustainable mobility. This study examined the factors influencing perceived safety among users of different transport modes using the Transport Safety dataset analyzed with the R programming language.

The study involved data cleaning, preprocessing, exploratory data analysis, visualization, and inferential statistical analysis. Variables such as transport mode, scenario type, pavement condition, obstacles, vehicle traffic, bicycle traffic, pedestrian traffic, gender, and age were examined. Descriptive statistics, frequency distributions, histograms, boxplots, correlation analysis, independent samples t-test, Welch one-way ANOVA, Games-Howell post hoc test, and multiple linear regression were employed to analyze the data.

The findings revealed that gender significantly influenced perceived safety, with females reporting slightly higher safety ratings than males ($t(5769) = -4.13, p < 0.001$), although the effect size was small (Cohen's $d = -0.11$). Significant differences were also found among transport modes (Welch's $F(3, 3187.7) = 510.87, p < 0.001$). Walking recorded the highest perceived safety ratings, while E-Scooter users reported the lowest. Post hoc Games-Howell comparisons confirmed significant differences between almost all transport modes except between Walking and Car.

Multiple linear regression showed that transport mode, scenario type, pavement condition, obstacles, gender, and some age categories were significant predictors of perceived safety. The regression model explained approximately 36.3% of the variation in perceived safety (Adjusted $R^2 = 0.361, F(16, 5754) = 204.5, p < 0.001$). Vehicle traffic had a small negative effect on perceived safety, whereas bicycle traffic was not statistically significant. Multicollinearity diagnostics indicated no serious correlation among the explanatory variables, with all Variance Inflation Factor (VIF) values close to one.

The study concludes that transport infrastructure and travel mode play important roles in shaping individuals' perceptions of safety. Improvements in road design, cycling infrastructure, pavement quality, and shared transport facilities can contribute to higher perceived safety and encourage sustainable transportation choices. These findings provide useful evidence for transport planners and policymakers seeking to improve urban mobility and road safety.

INTRODUCTION

1.1 Background of the Study

Transportation plays a vital role in economic development, social interaction, and the overall quality of life. Efficient transport systems enable people to access employment opportunities, education, healthcare services, and recreational activities. As urban populations continue to grow, cities around the world are increasingly encouraging sustainable transport modes such as walking, cycling, and electric scooters to reduce traffic congestion and environmental pollution.

Despite these advantages, safety remains one of the major concerns affecting people's choice of transport mode. Safety in transportation can be viewed from two perspectives: objective safety, which refers to measurable indicators such as accident and injury rates, and perceived safety, which refers to how safe individuals feel while using a particular transport environment. Research has shown that people's perceptions of safety often influence travel behaviour more strongly than actual accident statistics.

Several environmental factors influence perceived transport safety. These include traffic volume, pavement quality, pedestrian facilities, cycling infrastructure, crossing facilities, and the presence of physical obstacles. Personal characteristics such as gender and age may also shape individuals' perceptions of safety. Understanding these factors is essential for designing transport systems that encourage active and sustainable mobility.

With the increasing availability of open datasets and statistical software such as R, researchers are able to analyze transport safety data more effectively. This study applies statistical techniques including exploratory data analysis, hypothesis testing, analysis of variance, and multiple linear regression to investigate factors influencing perceived transport safety.

1.2 Problem Statement

Many urban planners invest heavily in transport infrastructure with the expectation that improved facilities will encourage sustainable transport. However, people may still avoid walking, cycling, or using micro-mobility options if they perceive these environments as unsafe.

Although previous studies have examined transport safety, there remains a need to understand how different transport modes, demographic characteristics, and environmental conditions jointly influence perceived safety. Furthermore, applying statistical methods using R programming provides valuable insights that can support evidence-based decision-making in transportation planning.

This study therefore seeks to identify the factors that significantly influence perceived transport safety using a publicly available transport safety dataset.

1.3 Objectives of the Study

General objective: The aim of this study is to investigate factors influencing perceived transport safety using statistical analysis in R programming.

Specific Objectives:

1. describe the characteristics of the transport safety dataset using descriptive statistics;
2. examine the distribution of perceived safety ratings among different transport modes;
3. determine whether perceived safety differs significantly between male and female respondents;
4. investigate whether perceived safety differs among transport modes;
5. identify the environmental and demographic factors that significantly influence perceived safety using multiple linear regression.

1.4 Research Questions

- I. What is the overall distribution of perceived safety ratings?
- II. Does perceived safety differ between males and females?
- III. Does perceived safety differ among transport modes?
- IV. Which transport mode is perceived as the safest?
- V. Which demographic and environmental variables significantly predict perceived safety?

1.5 Significance of the Study

The findings of this study are expected to benefit several stakeholders.

Urban planners may use the findings to improve transport infrastructure that enhances perceived safety.

Policy makers may use the results to support sustainable transportation policies.

Researchers may use the study as a reference for future work involving transport safety and statistical modelling.

Students studying data science, statistics, and transportation engineering may use the project as an example of applying R programming to real-world datasets.

1.6 Scope of the Study

This study focuses on analysing perceived transport safety using a secondary dataset obtained from a publicly available source. The study examines the relationships between perceived safety and selected demographic variables, transport modes, and environmental characteristics.

The statistical analyses are performed using R programming language. Techniques employed include descriptive statistics, exploratory data analysis, independent samples t-test, Welch's ANOVA, Games–Howell post hoc analysis, correlation analysis, and multiple linear regression.

1.7 Concept of Perceived Safety

Perceived safety refers to an individual's subjective feeling of how safe or unsafe a transport environment or travel mode is. Unlike objective safety, which is measured using accident records or injury statistics, perceived safety reflects personal judgments influenced by infrastructure, traffic conditions, demographic characteristics, and previous experiences.

Perceived safety is an important determinant of transport behaviour because individuals often choose their preferred mode of transportation based on how safe they believe it is rather than the actual level of risk.

In this study, perceived safety was measured using a seven-point Likert scale where:

- i. 1 = Very Unsafe
- ii. 2 = Unsafe
- iii. 3 = Slightly Unsafe
- iv. 4 = Neutral
- v. 5 = Slightly Safe
- vi. 6 = Safe
- vii. 7 = Very Safe

The dataset records respondents' safety ratings after evaluating different transport scenarios.

1.8 Factors Influencing Perceived Safety

1.8.1 Transport Mode

Different transport modes expose users to different levels of interaction with vehicles and infrastructure.

Examples include:

- i. Walking
- ii. Private cars
- iii. Electric bicycles (E-Bikes)
- iv. Electric scooters (E-Scooters)

Research suggests that users of micro-mobility modes such as e-bikes and e-scooters often perceive greater exposure to road hazards because they share road space with motor vehicles.

1.8.2 Road Infrastructure

Road design plays a significant role in determining perceived safety.

Examples include:

- i. Sidewalk width
- ii. Presence of cycle lanes
- iii. Shared-space roads
- iv. Urban roads without cycling facilities

The dataset used in this project evaluates four infrastructure types:

1. Urban road with sidewalks less than 1.5 m
2. Urban road with sidewalks at least 1.5 m
3. Urban road with cycle lane
4. Shared space with a 30 km/h speed limit

1.8.3 Traffic Characteristics

Traffic conditions also influence safety perceptions.

These include:

- i. Vehicle traffic density
- ii. Bicycle traffic density
- iii. Pedestrian density

High vehicle traffic often reduces perceived safety, whereas dedicated cycling facilities and pedestrian-friendly environments tend to improve it.

1.8.4 Road Environment

Environmental conditions influence perceived safety.

Examples include:

- i. Pavement quality
- ii. Road obstacles
- iii. Zebra crossings
- iv. Signal-controlled pedestrian crossings

Good infrastructure generally increases safety perception.

1.8.5 Demographic Characteristics

Previous studies indicate that demographic characteristics influence transport choices and safety perception.

Examples include:

- i. Gender
- ii. Age
- iii. Education
- iv. Employment status
- v. Income
- vi. Vehicle ownership

Women generally report higher concern for safety than men, while older adults often perceive greater transport risks.

METHODOLOGY

2.1 Research Design

This study employed a **quantitative research design** using secondary data obtained from a publicly available transport safety dataset. Quantitative research was selected because it allows statistical analysis of numerical and categorical data to identify relationships between variables and test hypotheses objectively.

The study focuses on understanding the factors that influence individuals' perceived safety when using different transport modes. Statistical techniques were employed to describe the data, compare groups, and develop predictive models.

2.2 Data Source

The dataset used in this study (Tzouras, P.G., Pastia, V., Kaparias, I. *et al.* Exploring the effect of perceived safety in first/last mile mode choices. *Transportation* **52**, 2145–2183 (2025).) is the **Rating Safety** dataset, which contains information on individuals' perceptions of transport safety under different transport scenarios.

The original dataset consisted of:

- i. 5,771 observations
- ii. 31 variables

The dataset includes variables relating to:

- i. Transport mode
- ii. Perceived safety ratings
- iii. Road infrastructure
- iv. Traffic conditions
- v. Demographic characteristics
- vi. Vehicle ownership
- vii. Transport frequency

The data was imported into **R Studio** for cleaning, exploration and statistical analysis.

2.3 Software Used

The analysis was conducted using **R Programming Language** within **RStudio**.

2.1: R Packages Used

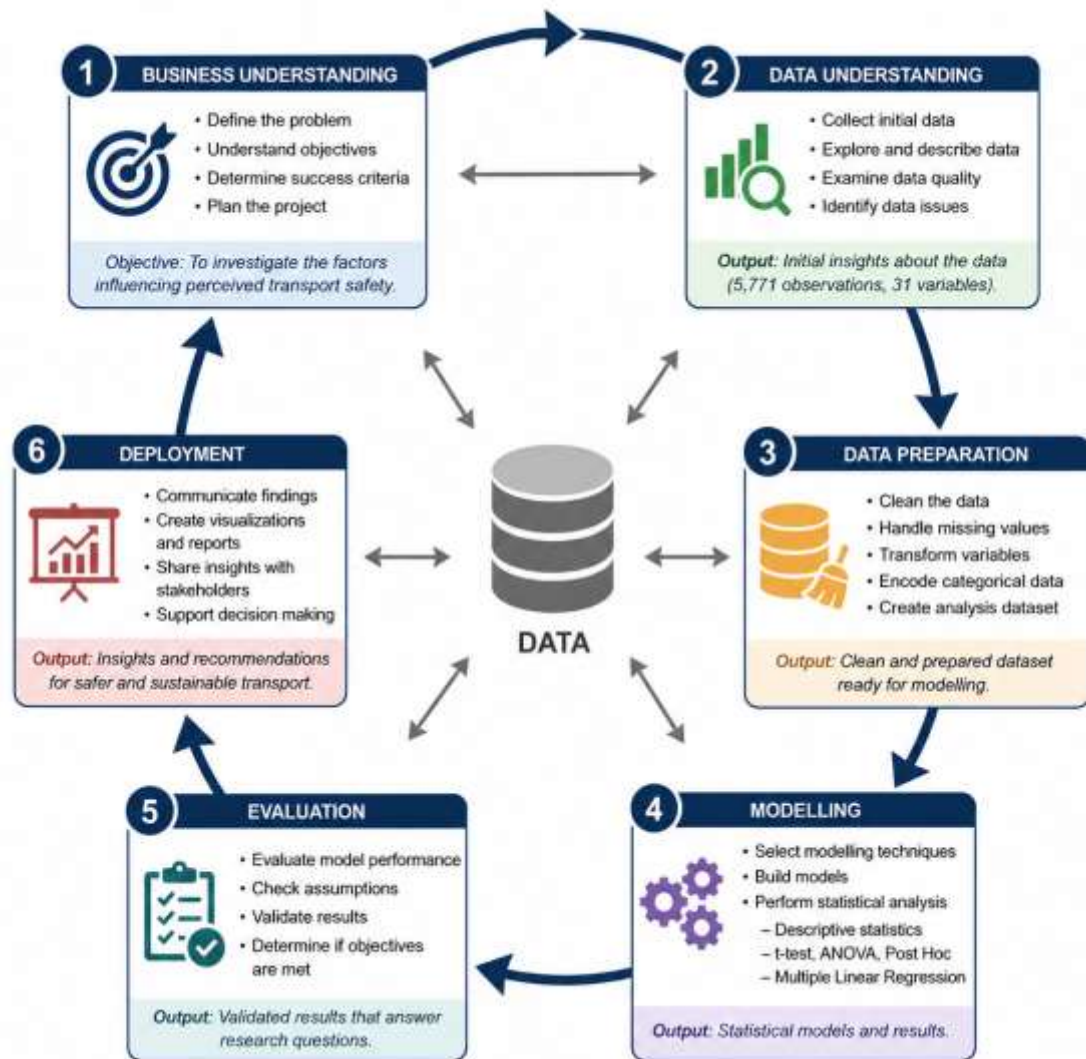
Packages	Purpose
tidyverse	Data manipulation
dplyr	Data cleaning and transformation
psych	Descriptive statistics
ggplot2	Data visualization
plotly	Interactive visualization
corrplot	Correlation plots
car	Levene's Test and VIF
rstatix	Statistical tests and effect sizes

2.4 Research Methodology

This study adopted the **Cross-Industry Standard Process for Data Mining (CRISP-DM)** methodology.

CRISP-DM provides a structured framework for carrying out data mining and statistical analysis projects. The methodology consists of six iterative phases:

1. Business Understanding
2. Data Understanding
3. Data Preparation
4. Modelling
5. Evaluation
6. Deployment



2.1: illustrates the CRISP-DM framework adopted in this study.

2.4.1 Business Understanding

The first phase involved identifying the research problem and defining the study objectives.

The study aimed to investigate the factors influencing perceived transport safety among different transport modes.

The objectives of the study were to:

- i. examine perceived safety across transport modes;
- ii. determine whether gender influences perceived safety;

- iii. investigate the effects of road infrastructure and traffic conditions on perceived safety; and
- iv. develop a regression model capable of predicting perceived safety.

These objectives guided all subsequent stages of the research.

2.4.2 Data Understanding

This phase involved exploring the dataset before any cleaning or transformation.

Activities carried out included:

- i. viewing the dataset;
- ii. examining the dimensions;
- iii. checking data types;
- iv. reviewing variable names;
- v. generating summary statistics;
- vi. identifying missing values; and
- vii. checking for duplicate observations.

The dataset contained both numerical and categorical variables representing transport characteristics and respondent demographics.

2.4.3 Data Preparation

Data preparation involved transforming the raw dataset into an analysis-ready dataset.

The following preprocessing procedures were performed:

i. Renaming Variables

Several variables were renamed to improve readability.

Examples include:

2.2: The renamed variables

Original variable	New variable
pid	participant_id
tmode	transport_mode
psafe	perceived_safety
type	scenario_type
veh	vehicle_traffic
bike	bicycle_traffic
ped	pedestrian_traffic
pav	pavement
obst	obstacles

ii. Handling missing values

The variables containing missing observations included:

- i. Income
- ii. Bike frequency
- iii. E-scooter frequency
- iv. Public transport frequency

- v. Metro use frequency

The missing values represented non-responses rather than data entry errors and were therefore retained where appropriate.

iii. Handling duplicates

iv. Data Type Conversion

Several coded numerical variables were converted into categorical factors for easier interpretation.

Examples include:

- i. Transport mode
- ii. Gender
- iii. Scenario type
- iv. Pavement condition
- v. Obstacles
- vi. Crossing type
- vii. Age
- viii. Education
- ix. Employment
- x. Income
- xi. Vehicle ownership
- xii. Transport frequency

```

32 # Handling missing values
33 colSums(is.na(transport_safety))
34 sum(is.na(transport_safety))
35
36 # Checks for duplicates
37 sum(duplicated(transport_safety))
38
39 # Renaming some of my columns
40 library(dplyr)
41
42 transport_safety <- transport_safety %>%
43   rename(
44     participant_id = pid,
45     transport_mode = tmode,
46     perceived_safety = psafe,
47     scenario_type = type,
48     pavement = pav,
49     obstacles = obst,
50     crossing_type = cross,
51     vehicle_traffic = veh,
52     bicycle_traffic = bike,
53     pedestrian_traffic = ped,
54     scenario_type1 = type1,
55     scenario_type2 = type2,
56     scenario_type3 = type3,
57     scenario_type4 = type4,
58     crossing_option1 = cross1,
59     crossing_option2 = cross2
60
61
62
63
64
65
66
67
68
69
70 # Converting the variables into factors
71 transport_safety$transport_mode <- factor(
72   transport_safety$transport_mode,
73   levels = c("car", "ebike", "escoot", "walk"),
74   labels = c("Car", "E-Bike", "E-Scooter", "Walking"))
75
76 view(transport_safety)
77
78 class(transport_safety$transport_mode)
79
80 transport_safety$gender <- factor(
81   transport_safety$gender,
82   levels = c(0,1),
83   labels = c("Male", "Female"))
84
85 class(transport_safety$gender)
86
87 transport_safety$scenario_type <- factor(
88   transport_safety$scenario_type,
89   levels = c(1,2,3,4),
90   labels = c(
91     "Urban Road (<1.5m Sidewalk)",
92     "Urban Road (>=1.5m Sidewalk)",
93     "Urban Road with Cycle Lane",
94     "Shared Space (30 km/h)"
95   ))
96

```

2.2: Snippet of the code showing handling missing and duplicates values and also factor conversion

2.5 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to summarize the characteristics of the dataset and identify patterns.

The following descriptive statistics were generated:

- i. Mean
- ii. Median
- iii. Standard deviation
- iv. Frequency distributions
- v. Percentages

Visualizations produced included:

- i. Histograms
- ii. Bar charts
- iii. Boxplots
- iv. Correlation matrix

EDA provided an understanding of variable distributions before inferential analysis.

2.6 Descriptive Statistics

The numerical variables analysed included:

- i. Perceived Safety
- ii. Vehicle Traffic
- iii. Bicycle Traffic
- iv. Pedestrian Traffic

2.3: Descriptive Statistics of Numerical Variables

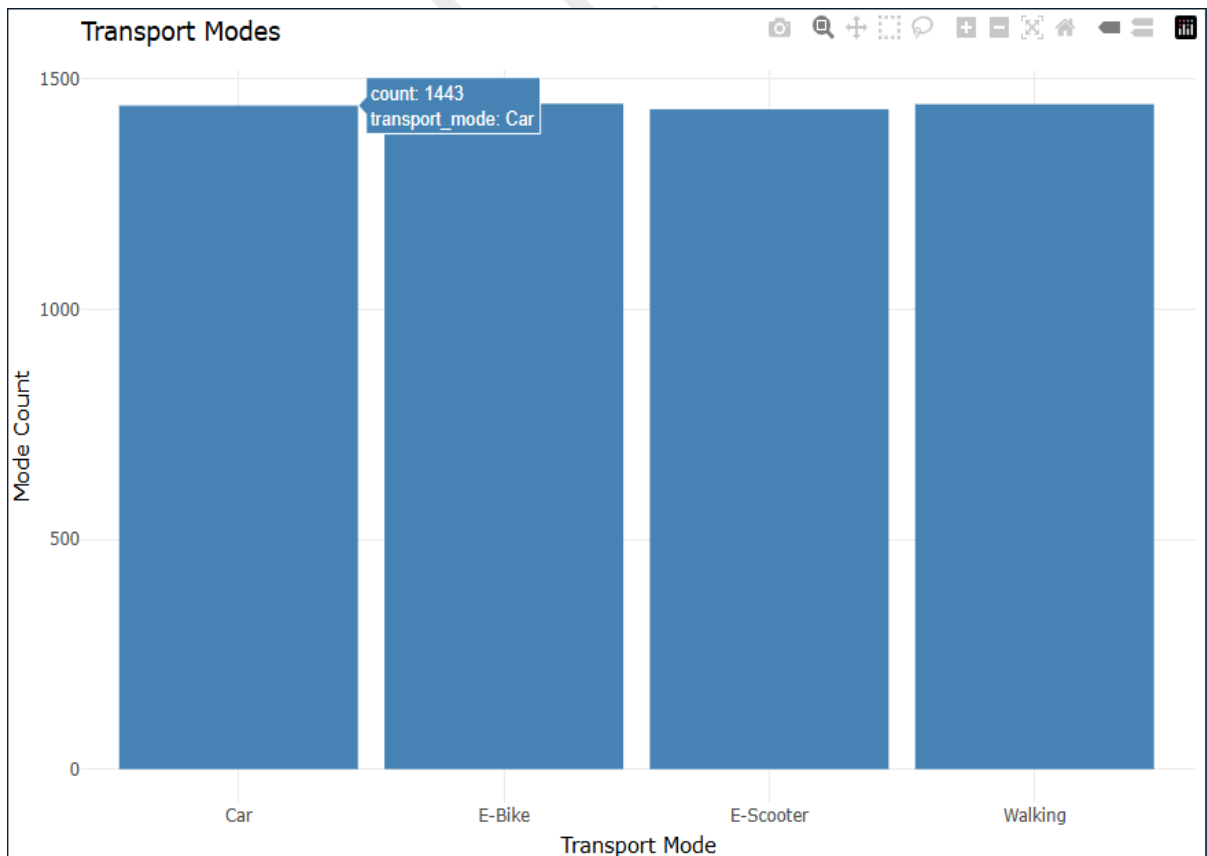
Variable	Mean	Median	Minimum	Maximum
Perceived Safety	4.38	5	1	7
Vehicle Traffic	60.0	60	20	100
Bicycle Traffic	50.0	50	10	90
Pedestrian Traffic	15.0	15	5	25

Interpretation

The average perceived safety rating was **4.38**, suggesting that respondents generally rated transport environments as moderately safe. Vehicle traffic averaged **60 vehicles**, while pedestrian traffic averaged **15 pedestrians**, indicating moderate traffic conditions across the evaluated transport scenarios.

2.7 Distribution of Respondents

2.7.1 Transport Mode



2.3: Transport Mode Bar Chart

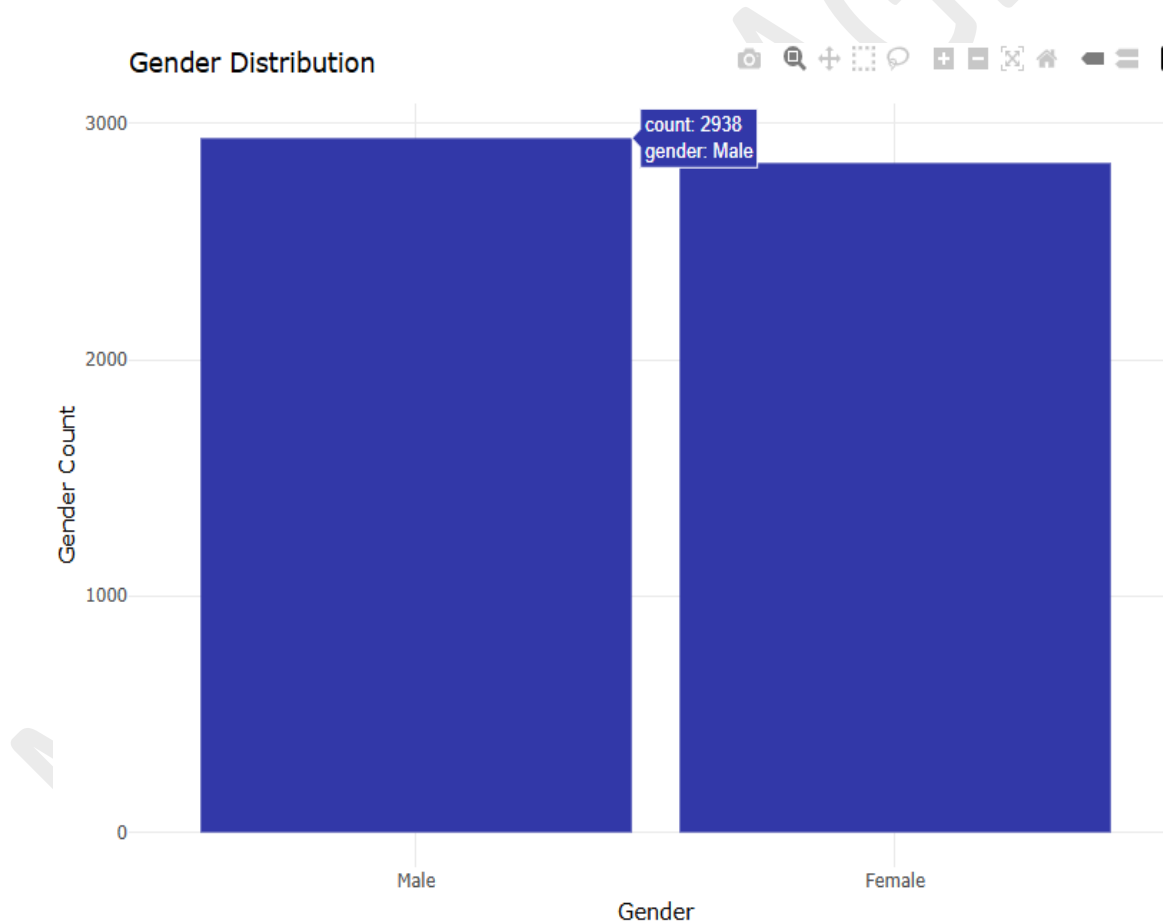
The dataset contained responses from four transport modes:

2.4: Transport Mode and their Counts

Transport Mode	Count
Car	1443
E-Bike	1447
E-Scooter	1435
Walking	1446

The distribution across transport modes was relatively balanced, allowing meaningful comparisons among the groups.

2.7.2 Gender Distribution



2.4: Gender Distribution Bar Chart

Table 1 2.5: Gender Distribution

Gender	Count
Male	2938
Female	2833

Both male and female respondents were represented in the dataset, allowing gender-based comparisons of perceived safety.

2.7.3 Distribution of Perceived Safety



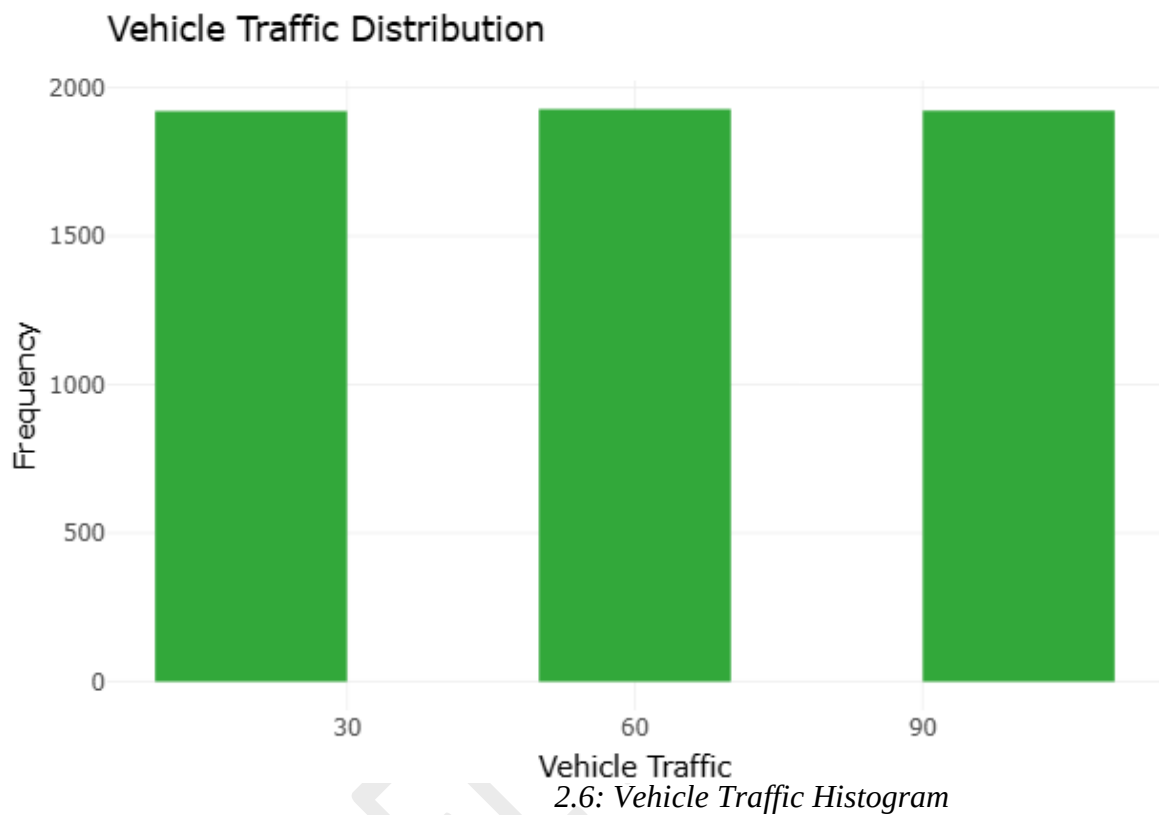
2.6: Histogram of Perceived Safety Ratings

2.6: Ratings and their Counts

Rating value	Rating Count
1	459
2	766
3	637
4	1004
5	957
6	1042
7	906

The histogram shows that perceived safety ratings ranged from 1 to 7, with most responses clustered around ratings of 4, 5, and 6. This suggests that respondents generally perceived transport environments as moderately to highly safe.

2.7.4 Vehicle Traffic Distribution



2.7: Vehicle Traffic and their Count

Vehicle Traffic	Count
20	1921
60	1927
100	1923

Vehicle traffic values were concentrated around the three predefined traffic levels (20, 60, and 100 vehicles), reflecting the controlled design of the transport scenarios used in the dataset.

2.8 Analysis

- i. Determine whether gender influences perceived safety.
Assumption Testing
 1. Levene's Test

2.8: Levene's Test

Test	p-value	Decision
------	---------	----------

Levene's Test	0.563	Equal variances assumed
---------------	-------	-------------------------

Since the p-value exceeded 0.05, the assumption of equal variances was satisfied.

2. Independent Samples t-Test

2.9: Gender and their Mean values

Gender	Mean
Male	4.28
Female	4.49

$$t = -4.132$$

$$p < 0.001$$

There was a statistically significant difference in perceived safety between male and female respondents ($t = -4.132$, $p < 0.001$). Female respondents reported higher perceived safety ($M = 4.49$) than male respondents ($M = 4.28$).

Effect Size of the Means of Gender

$$\text{Cohen's } d = -0.11$$

Although the difference was statistically significant, the effect size was **small**, indicating that gender had only a modest influence on perceived safety.

- ii. Determine whether perceived safety differs across transport modes.

Assumption Tests;

Shapiro-Wilk Test:

The Shapiro-Wilk test indicated that perceived safety scores were not normally distributed within each transport mode ($p < 0.001$).

However, because the sample size was large ($n = 5,771$), ANOVA was considered robust to violations of normality.

Levene's Test

Levene's Test produced:

$$F = 55.13$$

$$p < 0.001$$

The assumption of equal variances was therefore violated.

Welch ANOVA

2.10 Welch ANOVA Test

Test	p-value
------	---------

Welch ANOVA	<0.001
-------------	--------

Welch's ANOVA indicated significant differences in perceived safety among transport modes ($p < 0.001$).

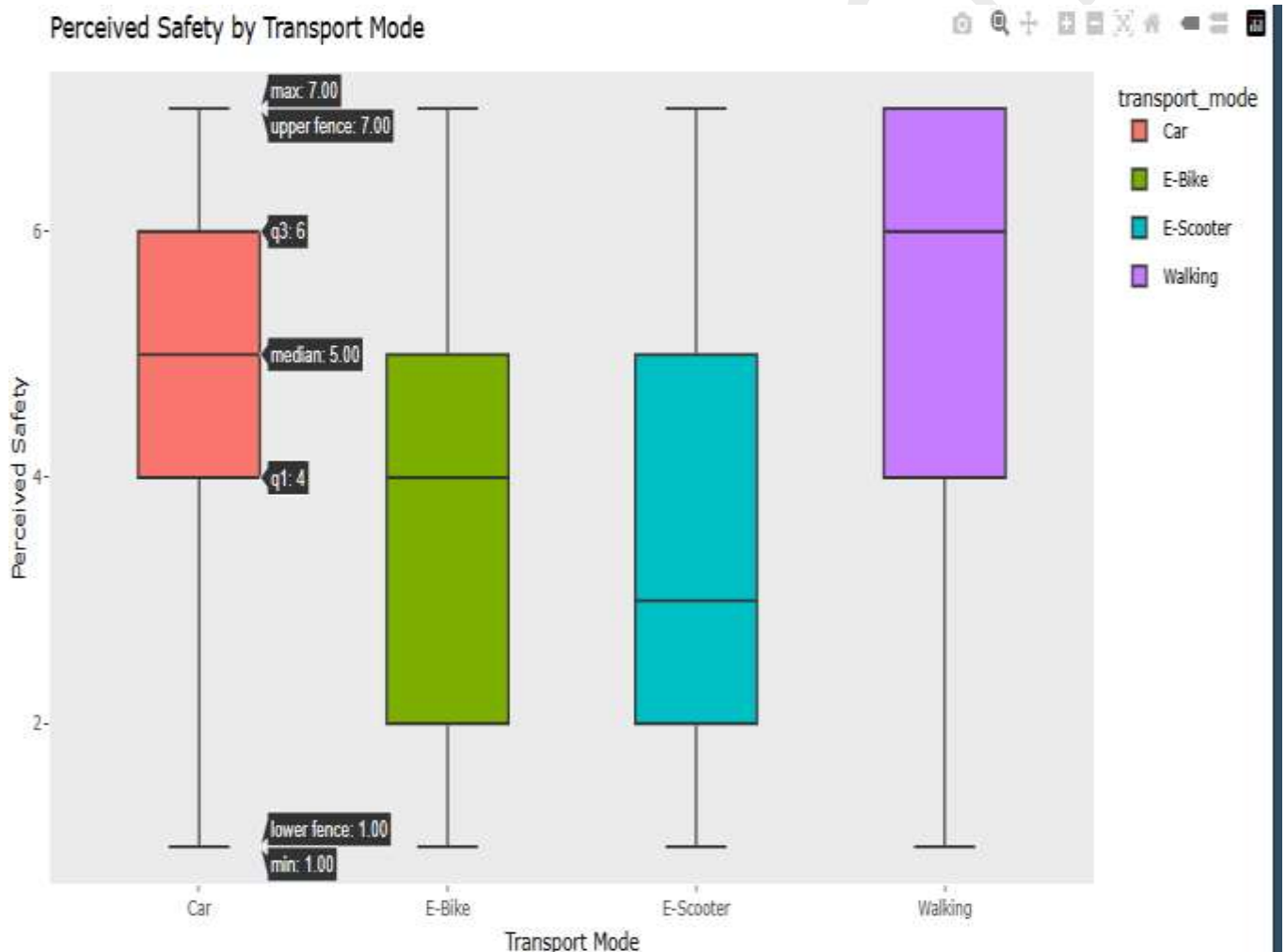
Effect Size

Eta-squared

$\eta^2 = 0.211$

Interpretation

Approximately 21.1% of the variation in perceived safety was explained by transport mode, representing a large practical effect.



2.7: Boxplot of Perceived Safety by Transport Mode

The boxplot shows that Car users had the highest median perceived safety score (5), while E-Bike users had the lowest median score (4). E-Bike ratings also displayed greater variability than Car ratings.

2.9 Multiple Linear Regression

A multiple linear regression model was developed to identify factors influencing perceived safety.

2.11: Multiple Linear Regression

Statistic	Value
R ²	0.363
Adjusted R ²	0.361
F-statistic	204.5
p-value	<0.001

The regression model was statistically significant and explained 36.3% of the variation in perceived safety. This indicates that the selected predictor variables provided a moderate explanation of respondents' safety perceptions.

Significant Predictors

Positive Predictors

- i. Urban Road (≥ 1.5 m Sidewalk)
- ii. Urban Road with Cycle Lane
- iii. Shared Space (30 km/h)
- iv. Good Pavement
- v. Many Obstacles
- vi. Female Gender
- vii. Walking

Negative Predictors

- i. E-Bike
- ii. E-Scooter
- iii. Higher Vehicle Traffic
- iv. Age 18–30
- v. Age 41–50
- vi. Age 51–65

2.10 Discussion of Findings

The results indicate that transport mode is a major determinant of perceived safety. Car users and pedestrians generally reported higher safety ratings than E-Bike and E-Scooter users. The large effect size ($\eta^2 = 0.211$) demonstrates that transport mode has substantial practical importance.

Gender also showed a statistically significant relationship with perceived safety, with female respondents reporting slightly higher safety perceptions than males. However, the small Cohen's d (-0.11) indicates that this difference is modest in practical terms.

The regression analysis further revealed that road infrastructure characteristics, including wider sidewalks, dedicated cycle lanes, and good pavement conditions, were associated with higher perceived safety ratings. Conversely, E-Bike use, E-Scooter use, and higher vehicle traffic levels were associated with lower perceived safety. The regression model explained 36.3% of the variability in perceived safety, suggesting that while the included variables are important, other factors not measured in the dataset may also influence safety perceptions.

3.1 Recommendations

Based on the findings, the following recommendations are proposed.

1. Improve cycling infrastructure

Since E-Bike and E-Scooter users reported the lowest perceived safety, city planners should invest in:

- i. Protected bicycle lanes
- ii. Dedicated scooter lanes
- iii. Improved lane markings

These measures may improve safety perceptions among vulnerable road users.

2. Improve pavement quality

Road authorities should prioritize:

- i. Maintenance of sidewalks
- ii. Repair of damaged pavements

- iii. Construction of wider pedestrian walkways

Improved pavement quality was positively associated with perceived safety.

3. Increase road safety awareness

Public education campaigns should encourage:

- i. Safe driving behaviour
- ii. Respect for pedestrians
- iii. Responsible cycling
- iv. Responsible E-Scooter use

This could improve both actual and perceived transport safety.

4. Improve urban road design

Local governments should incorporate:

- i. Signalized pedestrian crossings
- ii. Shared spaces
- iii. Dedicated cycling infrastructure
- iv. Traffic calming measures

into future road construction projects.

5. Encourage evidence-based transport planning

Policy makers should use data-driven approaches, such as statistical modelling and transport analytics, to guide transport planning decisions.

3.2 Limitations of the Study

Although the study achieved its objectives, several limitations should be acknowledged.

- i. The dataset was secondary data collected from a previous study.
- ii. The analysis relied on respondents' perceptions rather than actual accident records.

- iii. The regression model explained only **36.3%** of perceived safety, indicating that other influencing variables were unavailable.
- iv. Some demographic categories had relatively small sample sizes.
- v. Missing values in a few variables were removed during data preparation, which slightly reduced the sample size.

3.3 Conclusion

The study successfully identified important factors influencing perceived transport safety.

Transport mode emerged as one of the strongest determinants of perceived safety. Individuals travelling by Walking and Car generally felt safer than those using E-Bikes and E-Scooters.

Road infrastructure also played a significant role. Better pavement quality and improved road designs such as dedicated cycle lanes significantly increased perceived safety.

Female respondents reported slightly higher safety perceptions than males, although the practical difference was small.

The regression analysis demonstrated that transport infrastructure, road environment, demographic characteristics, and traffic conditions collectively explain approximately **36%** of perceived safety. This indicates that while these variables are important, additional psychological, environmental, and behavioural factors may also influence safety perceptions.

Overall, the study demonstrates that improving road infrastructure and designing transport systems that accommodate different road users can significantly improve public perceptions of transport safety.

This study demonstrates the usefulness of **R programming** and statistical modelling in analysing transport safety data. The findings highlight that transport mode,

infrastructure quality, and demographic characteristics significantly influence perceived safety. These insights provide valuable information for transport planners, policymakers, and urban designers seeking to create safer and more sustainable transportation systems.

References

1. Tzouras, P.G., Pastia, V., Kaparias, I. *et al.* Exploring the effect of perceived safety in first/last mile mode choices. *Transportation* **52**, 2145–2183 (2025). <https://doi.org/10.1007/s11116-024-10487-4>
2. https://zenodo.org/records/14967130?preview_file=rating_dataset_perceivedSafetyChoices.csv